

A Bone Glue with Sustained Adhesion under Wet Conditions

Laura Wistlich, Anja Rücker, Martha Schamel, Alexander C. Kübler, Uwe Gbureck,* and Jürgen Groll*

Bone glues often suffer from low adhesion to bone under wet conditions. This study aims to improve wet adhesiveness of a bone glue based on a photocurable poly(ethylene glycol) dimethacrylate matrix through in situ interpenetrating network formation by addition of six-armed isocyanate functional star-shaped prepolymers (NCO-sP(EO-*stat*-PO)). Biodegradable ceramic fillers are added to adjust the paste workability. The 3-point bending strength of the bone glues is in the range of 3.5–5.5 MPa and not significantly affected by the addition of NCO-sP(EO-*stat*-PO). Storage in phosphate buffered saline (PBS) decreases the bending strength of all formulations to approximately 1 MPa but the adhesion to cortical bone increases from 0.15–0.2 to 0.3–0.5 MPa after adding 20–40 wt% NCO-sP(EO-*stat*-PO) to the matrix. Bone glues without the NCO-sP(EO-*stat*-PO) additive lose their adhesiveness to bone after aging in PBS for 7 days, whereas modified glues maintain a shear strength of 0.18–0.25 MPa demonstrating the efficacy of the approach. Scanning electron microscopy and energy-dispersive X-ray spectroscopy investigations of the fracture surfaces prove a high amount of residual adhesive on the bone surface indicating that adhesion to the bone under wet conditions is stronger than cohesion.

properties of adhesives are required for short-time adhesion of bone fragments since subsequent osteosynthesis methods with plates and screws will be used to fix the larger bone pieces and to bear the load. In addition, successful bone adhesives should not act as regenerative barrier or interfere with fracture healing. Material approaches for bone adhesives are either based on natural adhesive compounds such as fibrin glue,^[4] mussel proteins,^[5] or polysaccharides^[6] or they are derived from synthetic compounds which are crosslinked in situ by a chemical reaction. The chemistry of the latter group includes cyanoacrylates,^[7] methacrylate monomers,^[8] polyurethanes,^[9] glass ionomer cements,^[10] or magnesium (calcium) phosphate cements.^[11]

Currently examined bone glues often suffer from a low adhesion to bone especially in the wet ambient conditions at the application site.^[2a] There are only few approaches to overcome this problem,

e.g., by using mussel adhesive proteins (MAPs) containing catechol moieties.^[5a] However, this often shows disadvantages because of possible immunogenic reactions and high costs for extracting the proteins from their host organisms.^[12] Coupling of the MAP-derived adhesion promoting amino acid 3,4-dihydroxyphenylalanine to poly(ethylene glycol) (PEG) in hydrogels results in minimal inflammatory response, but these hydrogels were demonstrated to serve only as soft tissue adhesive.^[13]

This study followed an alternative material approach by applying water soluble, isocyanate functional star-shaped prepolymers with ethylene oxide and propylene oxide statistically copolymerized in the backbone in a ratio of 4:1 (NCO-sP(EO-*stat*-PO)) as reactive additive to a poly(ethylene glycol) dimethacrylate (PEGDMA) matrix. Isocyanate-based polyurethane adhesives are widely used in technical applications, especially when wet material surfaces (e.g., wood) have to be fixed together.^[14] The NCO-sP(EO-*stat*-PO) additives have - in contrast to their counterparts from technical applications - a comparable low cytotoxicity in vitro and in vivo^[15] due to their high molecular weight where the NCO groups at the distal endings of the polymer chain only account for 2 wt% of the whole polymer. In water, these molecules cross-link to a three-dimensional polymer network through hydrolysis of their isocyanate groups followed by urea bridge formation between the prepolymers in a polyaddition reaction.^[16] The matrix was further

1. Introduction

The treatment of highly comminuted bone fractures with the formation of a certain number of bone fragments requires the use of adequate surgical techniques like nailing or plate osteosynthesis.^[1] Although screws and plates are currently available for fragment sizes of about 1 cm, smaller bone pieces are highly demanding for alternative fixation techniques such as in situ cross-linkable bone adhesives.^[2] The latter would offer the advantage of a consistent load transfer between the fracture surfaces.^[3] For clinical application, only moderate mechanical

L. Wistlich, A. Rücker, M. Schamel, Prof. U. Gbureck, Prof. J. Groll
Department for Functional Materials in
Medicine and Dentistry and
Bavarian Polymer Institute (BPI)
University of Würzburg
Pleicherwall 2, 97070 Würzburg, Germany
E-mail: uwe.gbureck@fmz.uni-wuerzburg.de;
juergen.groll@fmz.uni-wuerzburg.de
Prof. A. C. Kübler
Department of Cranio-Maxillo-Facial Surgery
University of Würzburg
Pleicherwall 2, 97070 Würzburg, Germany



DOI: 10.1002/adhm.201600902

modified by supplementation with ceramic fillers such as struvite ($\text{MgNH}_4\text{PO}_4 \cdot 6\text{H}_2\text{O}$), newberyite ($\text{MgHPO}_4 \cdot 3\text{H}_2\text{O}$), or gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$), which are thought to increase porosity following implantation due to their comparatively high solubility and should in addition act beneficial through the released ions to support ingrowth of new bone material.^[17]

2. Results and Discussion

Since many currently known material approaches for bone adhesives fail due to an insufficient bonding to wet and contaminated bone surfaces, the aim of this study was predominantly focused on improving the hydrolytic stability of the bone–adhesive interface. This was achieved by using isocyanate-terminated star-shaped prepolymers (NCO-sP(EO-stat-PO)) as additional component to a photocurable matrix. Isocyanates generally covalently react with protic functional groups such as thiols, amines, and hydroxyl groups in an addition reaction.^[16a] The reaction with water results in amines which react much more rapidly with remaining isocyanates in a second step, resulting in a two-step cross-linking reaction. This polyaddition occurs in parallel to the light induced radical polymerization of PEGDMA such that an interpenetrating network (IPN) structure results where the NCO groups are expected to additionally facilitate bone adhesion through partial reaction with protic groups at the bone surface.

2.1. Compositional Changes during Adhesive Setting and Immersion in Phosphate Buffered Saline (PBS)

In order to analyze both the polymerization reaction of methacrylate moieties and the hydrolysis and crosslinking of isocyanate groups, the adhesives were measured before and after irradiation and storage in PBS by Fourier transform infrared (FT-IR) spectroscopy (Figure 1). The conversion degree of the methacrylic double bond (1635 cm^{-1}) during photopolymerization was assessed by comparison with the $\text{C}=\text{O}$ peak area at 1720 cm^{-1} as internal standard (Figure 1A). Exposure times of 120 s resulted in a conversion rate of 50–60% of the $\text{C}=\text{C}$ bond, whereby NCO-sP(EO-stat-PO) appeared to slightly decrease the conversion rate of the methacrylate group to 50% for the 40 wt% NCO-sP(EO-stat-PO) containing sample. This conversion rate is specified as adequate for dental resins based on bifunctional methacrylates.^[18] Furthermore, the reaction of isocyanate groups was observed by FT-IR analysis. In Figure 1B, it is clearly visible that the functional groups of NCO-sP(EO-stat-PO) react within a couple of minutes and no NCO peak could be detected in samples containing 40 wt% NCO-sP(EO-stat-PO) already after 1 h of storage in PBS. Similar results were obtained for the specimens with gypsum and struvite fillers (data not shown).

By means of X-ray diffractometry (XRD), the dissolution and possible phase changes of the ceramic fillers by storage in PBS were determined. The dissolution of the filler leads to pore formation which is a prerequisite for a more rapid degradation and ingrowth of blood capillaries and bone-forming cells.^[19] Figure 2 shows as an example newberyite filled adhesives, for

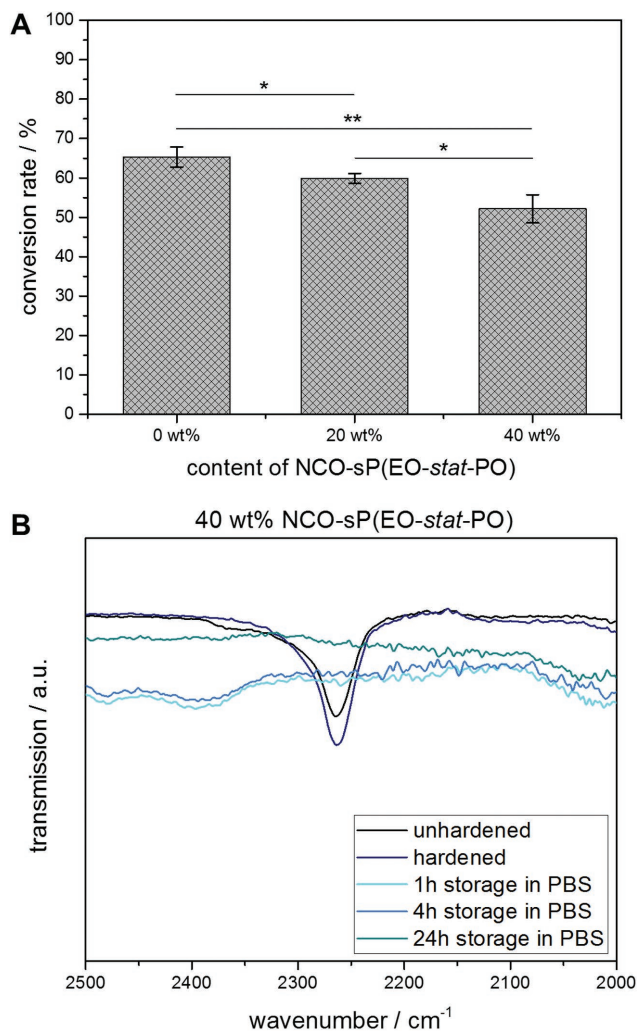


Figure 1. FT-IR measurements. A) Conversion rate of the methacrylate group in bone glues with newberyite filler and varying contents of NCO-bearing prepolymers. Level of significance of one-way analysis of variance (ANOVA): * means $p < 0.05$; ** means $p < 0.001$ ($n = 4$). B) NCO peak of the adhesives with 40 wt% NCO-sP(EO-stat-PO) and newberyite filler during the storage period of 24 h.

which the peak intensity decreased after 7 d in PBS due to partial filler dissolution. While adhesives with newberyite and struvite showed no changes of the phase composition, for gypsum, further crystalline compounds were observed by XRD and identified as sodium chloride, brushite, and octacalcium phosphate precipitates, which were formed by a dissolution–precipitation reaction in PBS (Figure S1, Supporting Information).

2.2. 3-Point Bending Strength

The bending strength with different ceramic fillers and its alteration by storing the samples in PBS were examined at day 0, 1, and 7 (Figure 3). Also the influence of the functional additive NCO-sP(EO-stat-PO) was analyzed by 3-point bending tests revealing the differences between samples without and with NCO-sP(EO-stat-PO). Reference samples

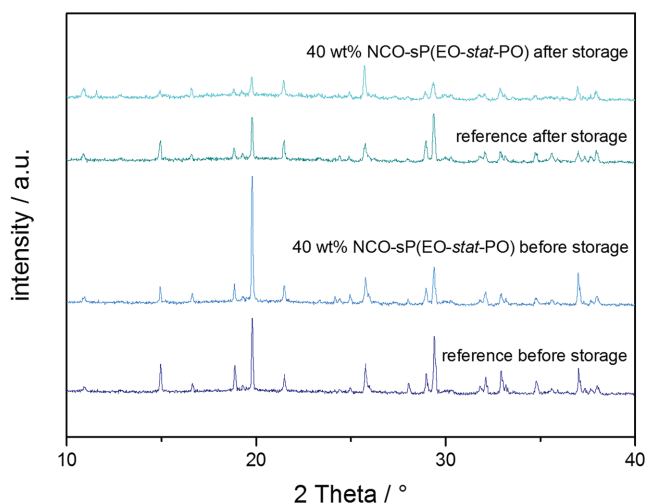


Figure 2. X-ray diffraction patterns of the samples with newberyite as filler before and after 7 d of storage in PBS. All diffraction peaks correspond to newberyite (powder diffraction file (PDF) No.: 35-0780).

without NCO-sP(EO-stat-PO) showed initially bending strengths between 3.45 ± 0.23 MPa (gypsum) and 5.28 ± 0.52 MPa (newberyite), which decreased to values between 0.5 and 1.0 MPa after storage in PBS for up to 7 d. This high strength loss can be explained by the absorption of water in the three-dimensional structure, which is created by the polymerization of PEGDMA^[20] and the cross-linking of NCO-sP(EO-stat-PO).^[16b,21] This swelling leads to a reduction of the mechanical properties accompanied with the dissolution of the ceramic filler. Modification of the matrices with 20 wt% NCO-sP(EO-stat-PO) decreased the bending strength for gypsum to values of 1.5 ± 0.4 MPa and for struvite to 3.4 ± 0.4 MPa. The same effect was observed for newberyite, also at higher additive loadings of 40 wt%. Aging of all NCO-sP(EO-stat-PO) adhesives in PBS resulted in bending strengths similar to the non-modified samples revealing no significant influence of the functional additive after storage.

2.3. Shear Bonding Strength on Bovine Bone

Independent of the type of ceramic material, the adhesive strength of the reference samples was increased by two to three times with the addition of 20 wt% NCO-sP(EO-stat-PO). While all reference samples without additive lost their adhesion to bone after storage for 24 h in PBS, the NCO-sP(EO-stat-PO) containing samples preserved at least a shear strength of 0.17 to 0.25 MPa after 7 d in PBS (Figure 4). The increase in strength values might be due to both the release of internal stresses of the glue which is created by shrinkage during polymerization and the increase of the contact area in an aqueous environment due to polymer swelling. A minimal required adhesive strength of bone glues was postulated as >0.2 MPa by Weber and Chapman,^[22] since a lower bonding strength is considered to be unsustainable under in vivo conditions.^[23] As mentioned above, the clinical need of such bone glues is the temporary adhesion of small bone fragments and not the long-time cohesion of the bone which can be established through external fixation.

While higher adhesion strengths to bone could be achieved by using other adhesives like cyanoacrylates^[24] or modified poly(methyl methacrylate) (PMMA) cements,^[25] these materials were tested commonly without wet storage or storage in an aqueous environment not exceeding 24 h. In our study, the glued constructs were stored for a time course of 7 d in PBS at body temperature without losing their adhesion strength in wet environment. Although adhesives based on ethyl or isobutyl cyanoacrylate^[26] can be stored for one month showing a higher bonding strength of 4.7 MPa, a clear disadvantage of these adhesives is both the toxicity of unreacted monomers and the formation of a rigid barrier between the bone fragments such they were considered to be not suitable for in vivo applications.^[27] Natural materials like fibrin adhesives showed very low strengths of about 0.01–0.2 MPa before and after storage in a wet environment.^[28] A more recent approach in literature is based on using copolymers together with phosphoserines derived from natural caddisfly adhesive silk to create a degradable system with bioresorbable components.^[29] Here, Bhagad et al. could increase the adhesive strength from 399 to 439 kPa by cross-linking with Ca^{2+} . Cohen et al. added ceramic fillers such as hydroxyapatite and tricalcium phosphate to a mixture of gelatin and alginate, which increased the bonding strength of soft tissue to bone from 8.4 to 18.1 kPa.^[30] Although this is in the range of the intrinsic strength of soft tissues, it is however far below the minimum required adhesive strength for bone glues. In general, comparing the data from different works is difficult since no standardized method exists and a wide variety of experimental setups and different surface pretreatments for the bone substrates (e.g., etching)^[31] were used. Furthermore, the bonding strength strongly depends on the type of bone (bovine or porcine; cortical or spongy)^[22,28] and on the environmental aging conditions (storage time, wet or moisture storage).^[2a,32]

2.4. Interface Analysis

Macroscopically, a cohesive failure was observed on specimens with 20 and 40 wt% NCO-sP(EO-stat-PO) content since significant material residues remained on the whole adhesion area after shear testing (Figure 5). For the reference sample without isocyanate addition (Figure 5A,D,F), no noticeable residues could be observed on the bone surface, which indicates an adhesive failure mode. Samples with 40 wt% NCO-sP(EO-stat-PO) (Figure 5B,C,E,G) showed more residues of the bone adhesive after the shearing test compared to 20 wt% NCO-sP(EO-stat-PO) samples with thicker layers and no visible bone within the adhesive area.

To obtain further information on the failure mode, the bone–adhesive interface was examined by scanning electron microscopy (SEM) (Figure 6), clearly demonstrating visible adhesive residues on the surface of samples with NCO-sP(EO-stat-PO) addition (Figure 6B) in contrast to the sample without additive (Figure 6A). This was confirmed by element mapping analyses using energy-dispersive X-ray spectroscopy (EDX) (Table 1, Figure 6C, and Figures S2 and S3, Supporting Information). Calcium (from bone hydroxyapatite) was predominantly found in adhesion areas when using adhesives without

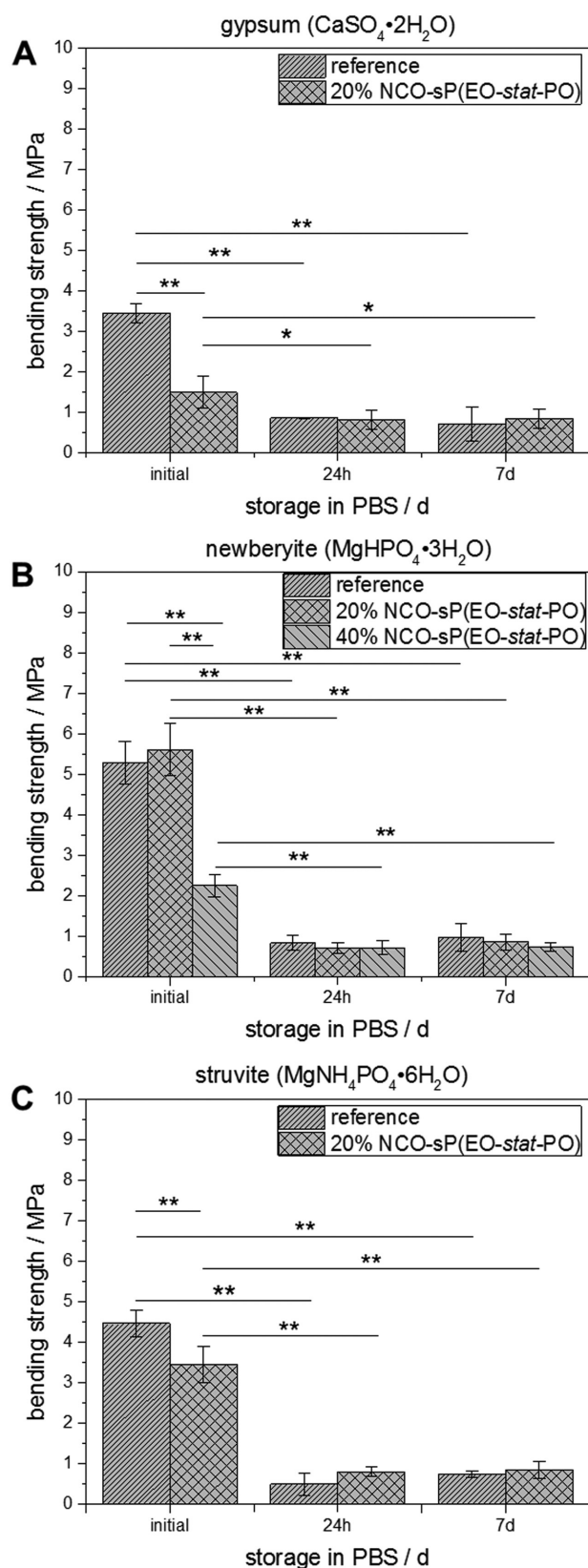


Figure 3. 3-point bending test of composites with different ceramic fillers and varying amounts of isocyanate bearing prepolymers. As reference, a sample without NCO-sP(EO-stat-PO) addition was used. Determination

NCO-sP(EO-stat-PO) (5.4 wt% in comparison to 0.1 wt% with additive). In contrast, the elemental composition of the adhesion area was dominated by magnesium from newberyite filler when using NCO-sP(EO-stat-PO) containing adhesives (3.4 wt% compared with 0.9 wt% without additive). This further confirmed a cohesive failure mode of the NCO-sP(EO-stat-PO) modified adhesives. The mass distribution of calcium for the samples without addition of NCO-sP(EO-stat-PO) indicated gaps in the material residues, which was also confirmed by SEM (Figure 6A). Due to the increased calcium levels, an enhanced adhesive failure was supposed compared to previously analyzed 40 wt% NCO-sP(EO-stat-PO) containing sample. Since all adhesives with isocyanate modification showed a cohesive failure mechanism within the adhesive, it is hence obvious that a further increase of the bonding strength of the whole system can be primarily achieved by a strength improvement of the matrix itself. The addition of mineral fillers such as gypsum, struvite, or newberyite in such bone glues is thought to enhance the degradation ability and to create porosity for an improved bone healing. This is a different concept to the works of Heiss et al.,^[33] in which adhesive degradation occurs by hydrolysis of the organic matrix while the filler β -tricalcium phosphate is mainly thought to neutralize acidic degradation products. All fillers from the current study have a higher solubility than β -tricalcium phosphate (0.39 mg L^{-1}) with struvite showing the lowest solubility (14.61 mg L^{-1}) followed by newberyite (232.83 mg L^{-1})^[17] and gypsum that can be dissolved in water up to $2\text{--}2.5 \text{ g L}^{-1}$ under equilibrium conditions.^[34] Although not experimentally demonstrated in this study, we believe that solubility of the fillers will lead to pore formation in a time frame of several weeks so that bone cells may subsequently grow into these voids. XRD results showed a change in phase composition and dissolution of minerals by storage in PBS. A clear disadvantage of gypsum was found to be the formation of lower soluble reprecipitates (mainly octacalcium phosphate and brushite) observed in XRD (Figure S1, Supporting Information). This phenomenon is a well-known problem of calcium phosphates and sulfates, which are thermodynamically unstable under in vivo conditions and transform into other phases in the presence of calcium and phosphate ions.^[35] Here, the use of magnesium phosphates is advantageous to avoid the formation of reprecipitates since magnesium ions are known to be an efficient inhibitor of hydroxyapatite crystal growth.^[36] This will in turn lead to a faster formation of porosity within the adhesive matrix and will hence achieve a better remodeling of the bone fracture. Indeed, XRD analysis of newberyite filler containing adhesives (Figure 2) showed no conversion into other crystal phases after 7 days.

2.5. Cytocompatibility

It is well known that low molecular weight isocyanate containing substances are often acutely toxic and harmful.^[37] This originates from the high reactivity of the isocyanates in

of bending strength was done initial and after storage in PBS for 1 and 7 d. The results are shown for the ceramic fillers A) gypsum, B) newberyite, and C) struvite. Level of significance of two-way ANOVA: * means $p < 0.05$; ** means $p < 0.001$ ($n = 6$).

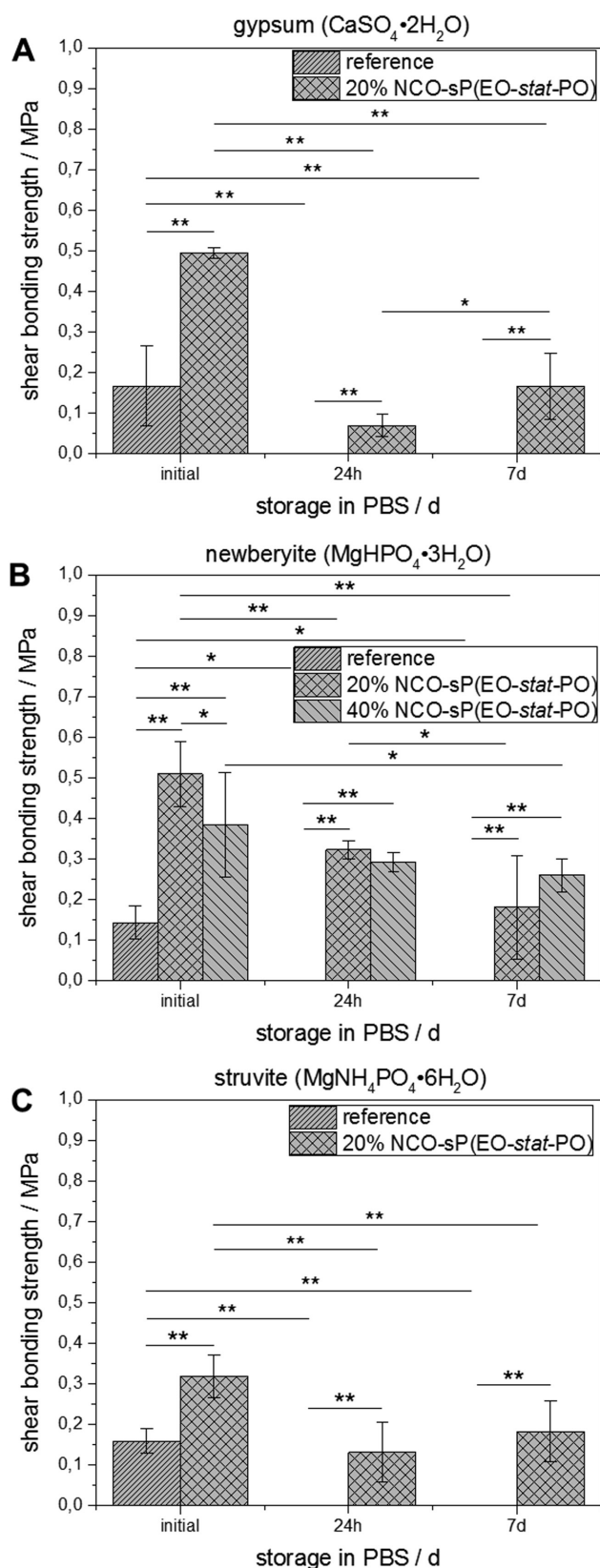


Figure 4. Shear bonding test of composites with different ceramic fillers and varying amounts of isocyanate groups. Bonding strength to bone was determined initially and after storage in PBS for 1 and 7 d. As reference, a

combination with the high mobility of low molecular weight compounds and their ability to enter cells. We thus used isocyanate functional prepolymers which have the advantage that the isocyanates only account for less than 2 wt% of the molecule. Furthermore, the mobility of the polymer with 12 kDa is lower than for small molecules, and the hydrophilic prepolymers do not easily penetrate cells. Most importantly, a rapid reaction of the isocyanates to amine groups and their subsequent cross-linking under the formation of biocompatible urea bonds will rapidly generate a three-dimensional polymer network in which the remaining isocyanate group will be immobilized and thus not be able to negatively affect the biological surrounding. This is of essential importance for potential use in humans. Initial cell tests using 1 wt% *N,N,N',N'*-Tetramethylethylenediamine (TEMED) failed with both L929 fibroblasts and hFOB 1.19 cell line even on samples without NCO-sP(EO-stat-PO), which could be attributed to the low cytocompatibility of the initiator TEMED.^[38] This problem was overcome by reducing TEMED concentration in the materials to 0.1 wt% for cell testing. As this material modification was proved to have either no or even a positive effect on the material mechanical performance and adhesion to bone, we prepared samples without TEMED, again without detrimental effects on bone adhesion (Figure 7). After storage in PBS for 1 and 7 d, the bending strength of all samples was in the range of 1 to 2 MPa (Figure 7A) with an adhesive strength to bone of 0.2–0.6 MPa (Figure 7B), where the maximal adhesive strength was reached with the TEMED free sample.

Investigations of the in vitro cytocompatibility of such modified bone adhesives with newberyite as filler according to DIN EN ISO 10993 (Figure 8A,B) showed only a small influence in cell number and activity for NCO-sP(EO-stat-PO) containing adhesives and hence proved the cytocompatibility of the materials according to ISO standard with more than 80% for cell number and activity compared to the reference. The hydrogels of pure NCO-sP(EO-stat-PO) even revealed higher cell count and activity which clearly demonstrated good cytocompatibility of the additive compound. A further cell test with the osteoblastic cell line hFOB 1.19 indicated that cells which were incubated with eluate from samples without or with functional additive showed a good or even better cell growth and activity compared to polystyrene (PS) (Figure 8C,D). The cell numbers of all specimens were above 80%, the activity values were much higher than PS for all samples. These results showed the cytocompatibility of the NCO-bearing prepolymers and demonstrated that TEMED is the more critical component in this context, with an additional negative effect on bone adhesion.

2.6. Applicability of the System

Apart from mechanical stability and adhesion to wet bone, further prerequisites for bone adhesives include an easy application by the surgeon with an adequate working time, a rapid

sample without NCO-sP(EO-stat-PO) addition was used. The results are shown for the ceramic fillers A) gypsum, B) newberyite, and C) struvite. Level of significance of two-way ANOVA: * means $p < 0.05$, ** means $p < 0.001$ ($n = 5$).

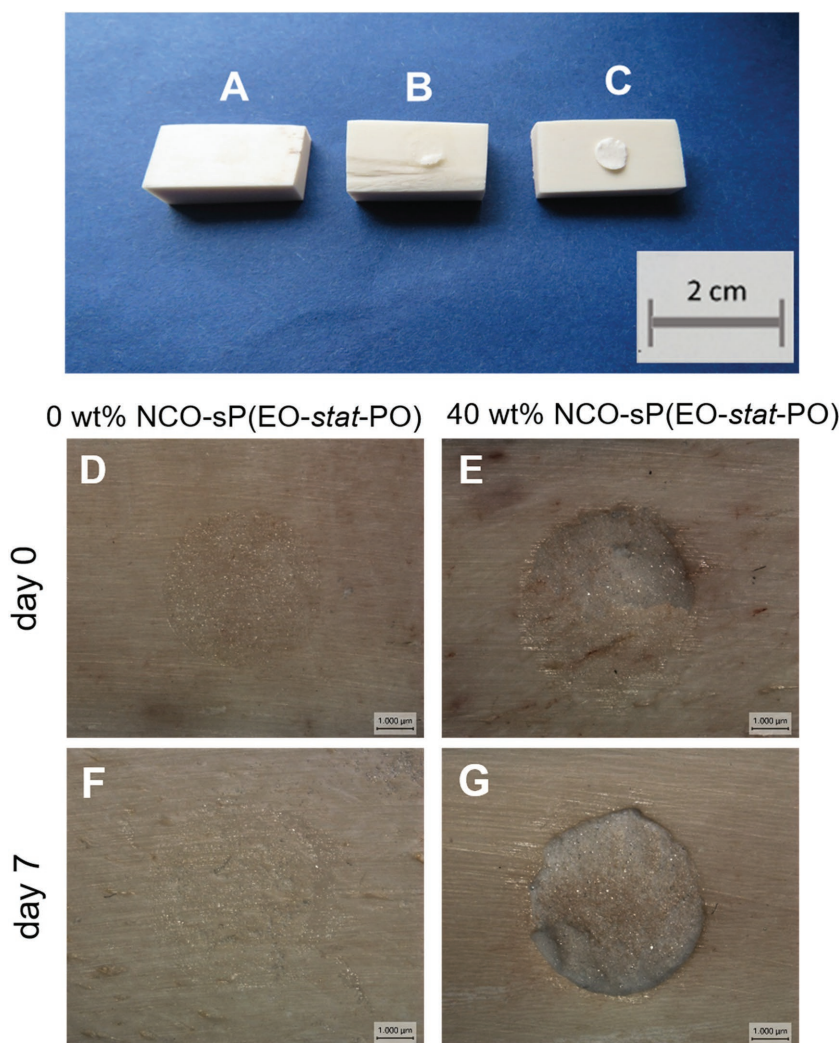


Figure 5. Interface analysis - macroscopic. Bone surfaces of adhesives with newberyite filler and varying contents of isocyanate groups after 7 d storage in PBS and shear testing. The adhesive areas of the A) reference sample, of a sample with B) 20 wt% NCO-sP(EO-stat-PO) and a sample with C) 40 wt% NCO-sP(EO-stat-PO) are shown as example. D, E) Interface images of samples with newberyite as mineral filler without addition of NCO-sP(EO-stat-PO) and F, G) with 40 wt% NCO-sP(EO-stat-PO) before and after 7 d of storage in PBS.

and low exothermic setting reaction which can preferably be triggered by external stimuli as well as manufacturing related issues such as shelf-life, sterilizability, and cost efficacy.^[2a] The adhesives from this study already meet several requirements for bone adhesives as mentioned above, e.g., they are ready to use and easy to handle and no additional mixing step is required prior to application to the bone fragments as revealed in an *ex vivo* application on a porcine jawbone fracture (Figure 9A,B). An advantage of using a light-curing adhesive compared to chemically hardened glues is a largely extended application time for a surgeon to manipulate and arrange bone fragments, followed by a hardening on demand after irradiation. The comparatively low viscosity enabled an easy flow at the application site and sufficient wetting of the bone fragment surfaces. The flowability was found to mainly depend on the type of filler rather than on the addition of functional NCO-bearing additive,

e.g., the use of gypsum or newberyite led to a pasty texture which exhibited good handling properties. In contrast, the paste with struvite as filler was macroscopically dryer and not free flowing and therefore difficult to apply. These observed effects are likely related to the different particle size distributions of the filler particles (data not shown) and an optimization of this parameter (e.g., by using bimodal size distribution) is thought to further improve the flowability of the paste. The type of filler also had a strong influence on the depth of curing of the adhesives. The curing depth corresponds to the depth of a fracture which could be treated by a surgeon with the developed adhesive by light curing from only one side.^[39] While gypsum filler resulted in a depth of curing of about 4.7 mm, the use of magnesium phosphates increased the depth to 6.9 mm (newberyite) and 10.9 mm (struvite) (Figure S4, Supporting Information). Since in all cases a similar amount of filler was added to the organic matrix, this is probably correlated to the optical properties of the filler particles.^[40] As demonstrated in Figure S4 (Supporting Information), the depth of curing from one side as it would be applied by the surgeon is sufficient to fix small bone fragments and hence further demonstrates the clinical applicability of our approach.

To verify an adequate shelf-life and sufficient working time of the investigated adhesives, it was demonstrated that pastes with all filler types without NCO-sP(EO-stat-PO) could be stored for at least 8 months under light protected conditions and remained workable for this time as proved by preparing cement disks which were as stable as those made of freshly mixed adhesives. However, pastes with NCO-sP(EO-stat-PO) additive hardened in the syringe within 48 h, which is likely a result of urea-bond

formation due to the presence of water within the pastes. This may either originate from the PEGDMA matrix itself or it might be slowly released from the hydrated ceramic fillers. Nevertheless, this problem could be solved by a separation of PEGDMA/filler pastes and NCO-sP(EO-stat-PO) in a double cartridge delivery system (MedMix, Switzerland), which enables mixing of both components directly before application in the extruder of the syringe (Figure 9C). Storage of the pastes in such a double cartridge system at 4 °C under light protection extended the shelf-life to at least 23 d. At this time point, the contents of the different cartridges were still paste-like and could be mixed and hardened similarly to freshly prepared materials. Working times for the pastes were found to be adequate for a clinical handling with 15 min for isocyanate-free samples because of polymerization by daylight. Samples with NCO-sP(EO-stat-PO) modification additionally

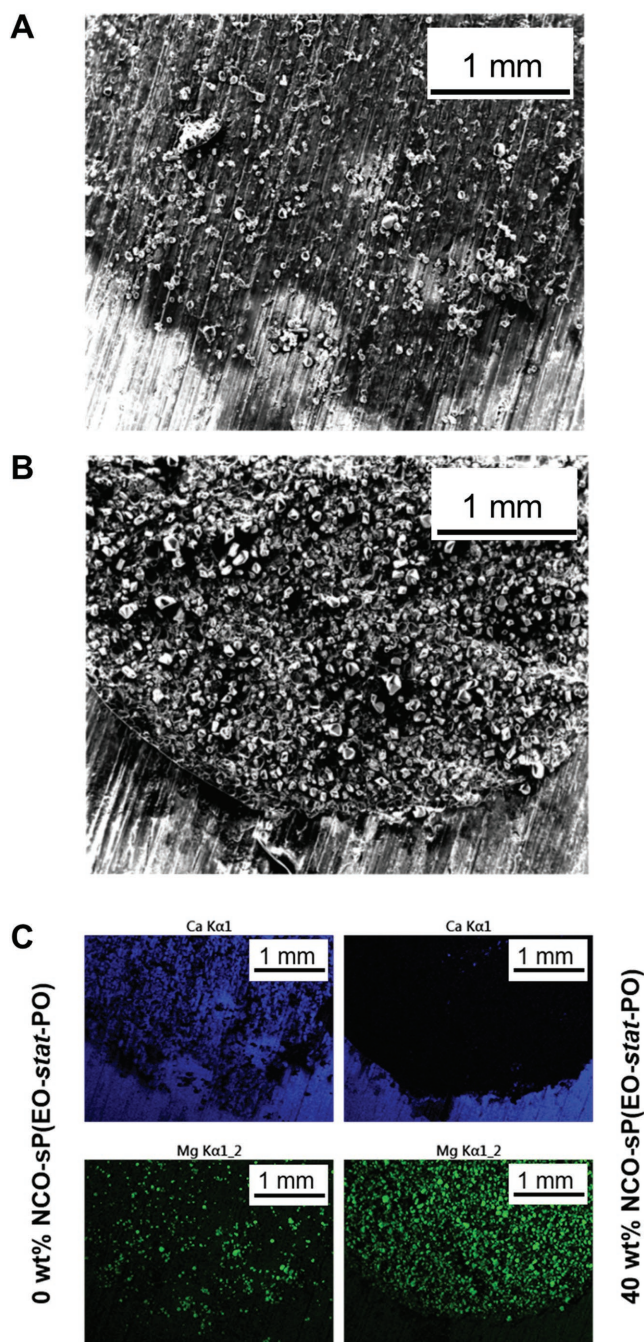


Figure 6. Interface analysis - microscopic. Scanning electron image of A) a 0 wt% and B) a 40 wt% NCO-sP(EO-stat-PO) containing sample with newberyite filler after storage for 7 d in PBS and subsequent shear testing. C) Element distribution analysis of the adhesive area on bone with newberyite as ceramic filler and with 0 wt% and 40 wt% NCO-sP(EO-stat-PO) addition.

hardened by exposure to air humidity because of hydrolysis to amine groups and subsequent crosslinking with NCO-groups such that the working time was reduced to 6 min. However, even this shorter working time is clinically acceptable and exceeds the working time for fibrin glues by an order of magnitude.^[2b]

Table 1. Element composition of adhesive points on bone without and with NCO-sP(EO-stat-PO) after shear testing. The images in Figure S2 (Supporting Information) show the corresponding topographic appearance of Kα1 for the respective elements.

Element [wt%]	0% NCO-sP (EO-stat-PO)	40% NCO-sP (EO-stat-PO)
Calcium (Ca)	5.4	0.1
Magnesium (Mg)	0.9	3.4
Carbon (C)	52.9	47.8
Phosphorus (P)	3.7	3.9
Oxygen (O)	37.1	44.2

3. Conclusion

In the current study, we transferred the improved adhesiveness of isocyanate containing technical glues under wet conditions to bone adhesives. This was achieved by employing isocyanate functional hydrophilic prepolymers as additive

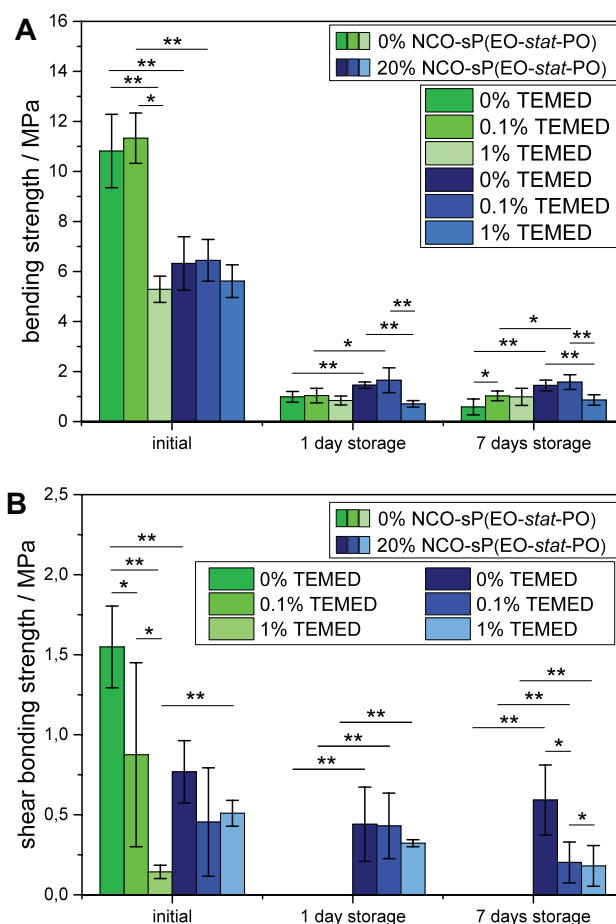
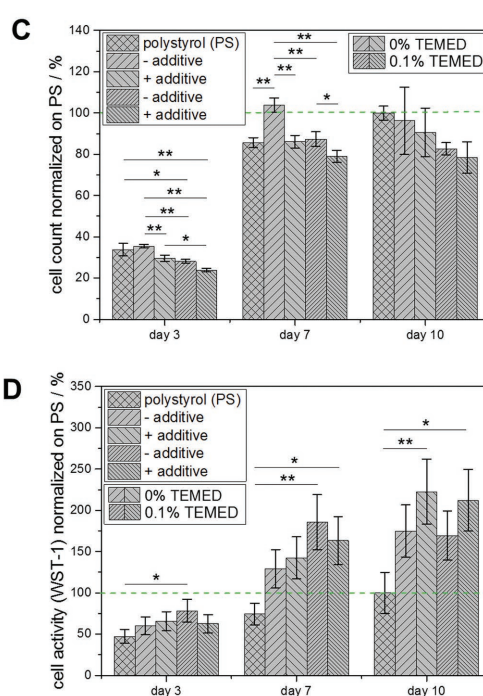
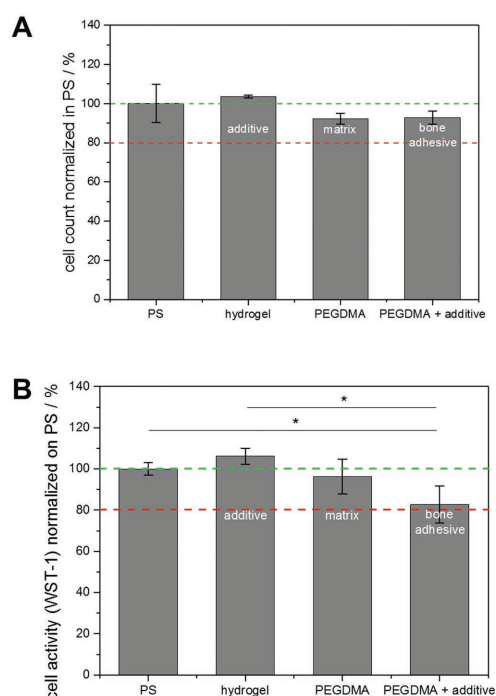


Figure 7. Mechanical testing with lower TEMED amount. A) Bending and B) bonding strength of newberyite samples without and with NCO-sP(EO-stat-PO) with different TEMED concentrations (0, 0.1, and 1 wt%). Level of significance of three-way ANOVA: * means $p < 0.05$, ** means $p < 0.01$ ($n = 6$ for bending strength and $n = 5$ for shear bonding strength). The statistical significance is indicated between different TEMED concentrations and different NCO-sP(EO-stat-PO) contents, not between time points.

fibroblasts L929



osteoblasts hFOB 1.19

Figure 8. Cytocompatibility. A) Cell number and B) cell activity of L929 fibroblasts according to WST-1 test of a 20 wt% NCO-sP(EO-*stat*-PO) hydrogel and bone adhesives with and without 20 wt% NCO-sP(EO-*stat*-PO) (stated as additive) using newberyite as filler according to DIN EN ISO 10993-5. Samples were prepared with 0.1 wt% TEMED. Cells were cultured in 100% DMEM eluate from the samples for 48 h. C) Cell number and D) cell activity of hFOB 1.19 according to WST-1 test of bone adhesives with and without 20 wt% NCO-sP(EO-*stat*-PO) with newberyite as filler using eluate testing. Samples were prepared with 0 wt% or 0.1 wt% TEMED. Cells were cultured in 100% DMEM eluate from the samples for 3, 7, and 10 d. Level of significance of one-way and three-way ANOVA: * means $p < 0.05$, ** means $p < 0.01$ ($n = 3$). The statistical significance is indicated between different TEMED concentrations and different NCO-sP(EO-*stat*-PO) contents, not between time points.

to a photochemically crosslinked PEGDMA matrix and thus forming an IPN structure. This approach significantly increased the bonding ability to bone even after storage in PBS for up to 7 days. We demonstrate cytocompatibility of the system and show its straightforward practical applicability.

Although the bone adhesives developed in this work meet the demand for a minimum adhesive strength of 0.2 MPa in a wet environment, they are still nonoptimized systems, for example with regard to the IPN formation between the network forming methacrylate matrix and the adhesion promoting urethane matrix.^[41] We suppose that a chemical cross-linking of both networks, e.g., by combination of both functionalities (methacrylate and isocyanate) in one monomer, will be helpful to further increase both the inherent chemical stability as well as the adhesion to wet bone surfaces. In addition, the introduction of calcium (magnesium) chelating groups (such as bisphosphonates) into the organic network is thought to strengthen the interface between filler particles and polymer matrix to further improve mechanical properties and adhesive strength to bone. Also the degradation of the organic matrix that so far relies on the hydrolysis of methacrylate ester groups might be improved by using resorbable polylactides and polyglycolides which have a free terminal hydroxyl group for introduction of cross-linkable methacrylate groups. This was previously demonstrated in lung and dental technology^[2b] showing a good degradation rate and biocompatibility.^[3] Ho and Young investigated the degradation of poly(methacrylic-*co*-lactic) acid and could

reveal that a higher number of lactic units led to an increased mass loss over 14 weeks; adhesives with more propylene glycol units required longer curing but showed a slower hydrolytic degradation.^[42] Nonetheless, we believe that this study opens a new strategy toward cytocompatible and degradable bone glues that do not create a barrier but allow for bone formation upon fracture healing.

4. Experimental Section

Ceramic Filler Particles: $\text{Mg}_3(\text{PO}_4)_2$ was synthesized by mixing $\text{MgHPO}_4 \cdot 3\text{H}_2\text{O}$ (107.0 g) and $\text{Mg}(\text{OH})_2$ (17.09 g) (both Sigma-Aldrich, Steinheim, Germany) in a planetary ball mill (PM 400 Retsch, Haan, Germany), followed by sintering for 5 h at 1050 °C in a furnace (Oyten Thermotechnik, Germany). The sintering cake was manually crushed with mortar and pestle and finally sieved to a particle size $< 355 \mu\text{m}$. Struvite (magnesium ammonium phosphate, $\text{MgNH}_4\text{PO}_4 \cdot 6\text{H}_2\text{O}$) was synthesized by stirring $\text{Mg}_3(\text{PO}_4)_2$ (26.39 g) and $(\text{NH}_4)_2\text{HPO}_4$ (19.89 g) in pure water (500 mL) for 48 h. The precipitation product was dried at 60 °C for several hours and the phase purity was verified by X-ray diffraction analysis (XRD) with small amounts of farringtonite ($\text{Mg}_3(\text{PO}_4)_2$) and newberyite ($\text{MgHPO}_4 \cdot 3\text{H}_2\text{O}$) as by-products. Magnesium hydrogen phosphate trihydrate ($\text{MgHPO}_4 \cdot 3\text{H}_2\text{O}$, newberyite, $\geq 98\%$) and calcium sulfate dihydrate ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$, gypsum, 99%–101%) as other fillers were purchased from Sigma-Aldrich (Steinheim, Germany) and sieved $< 125 \mu\text{m}$ prior to use.

Bone Adhesive Preparation: Poly(ethylene glycol) dimethacrylate (5.5 g, 10 mmol) (PEGDMA, average M_n : 550; Sigma-Aldrich,

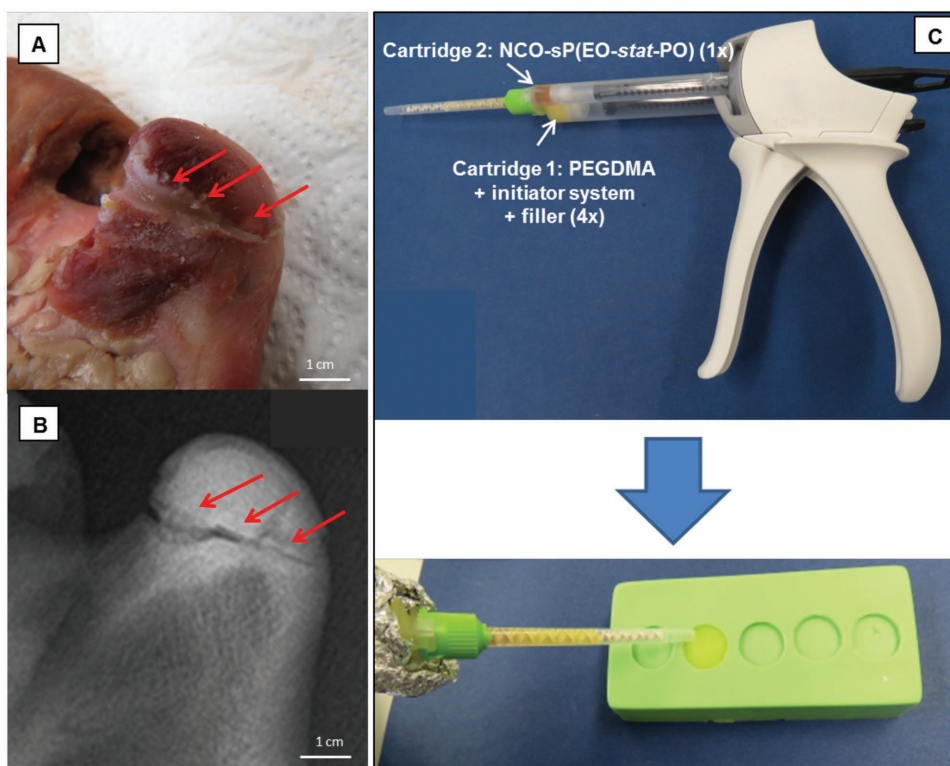


Figure 9. Ex vivo testing for demonstrating clinical applicability. A) Photograph and B) X-ray image of fixing of a porcine jawbone by using a bone adhesive with 20 wt% NCO-sP(EO-stat-PO). C) Bone adhesive filled in a double cartridge syringe: one cartridge: PEGDMA + initiator system + newberyite filler; second cartridge: NCO-sP(EO-stat-PO); mixing ratio: 4:1.

Steinheim, Germany) were mixed with D,L-camphorquinone (56.1 mg, 0.338 mmol) (Merck, Darmstadt, Germany) as radical initiator and N,N,N',N'-Tetramethylethylenediamine (72.4 μ L, 0.483 mmol) (TEMED, 99%; Sigma-Aldrich, Steinheim, Germany) as activator. Inorganic mineral fillers gypsum, newberyite, or struvite were added in a powder-liquid ratio of 1:1:1. For increasing the bonding strength to wet bone surfaces, 20 or 40 wt% of NCO-sP(EO-stat-PO) (DWI Leibniz-Institute for Interactive Materials, Aachen, Germany) were added to the organic phase before adding the ceramic filler.

Sample Preparation: Bone adhesive pastes were transferred into cuboid silicon rubber molds in order to prepare samples for 3-point bending strength testing (3 mm $\times$ 4 mm $\times$ 25 mm) and then crosslinked with an LED lamp (Bluephase100, Ivoclar Vivadent, Ellwangen, Germany). The test specimens were exposed to light three times for 30 s, then removed from the mold and irradiated for another 30 s from the bottom. Samples were tested either directly after curing or after storage in PBS for 24 h and 7 d. 3-point bending samples were tested in axial bending at a crosshead speed of 5 mm min⁻¹ using a static mechanical testing machine Zwick 1440 (Zwick, Ulm, Germany) and a 10 kN load cell. The bending strength was calculated by the device according to

$$\text{bending strength [MPa]} = \frac{3 F_{\max} * l}{2 * b * h^2} \quad (1)$$

where $F_{\max}$ is the failure load, l is the distance between the outer rolls, b is the width, and h is the height of the samples.

Preparation of Bone Disks and Adhesion Tests: Bovine bone from the femur was purchased from a butcher and used for adhesion tests. The diaphysis was cut with a bone saw into 2.5 cm thick slices, which were then divided into 0.5–1.0 cm thick pieces by means of a belt-saw Bizerba FK 22 (Balingen, Germany). The bone pieces were reworked to a final size of 1 cm $\times$ 2 cm $\times$ 0.5 cm by mechanical grinding and then stored in PBS buffer to prevent drying for maximum 5 d at 5 $^{\circ}$ C. The different

steps for preparing the samples for shear strength testing are shown in Figure S5 (Supporting Information). A hand-made device was used to build the adhesive dots for the shear testing. The bone disk was placed in a PMMA cavity and a silicone rubber matrix with a bore hole of 6 mm diameter and 5 mm height was placed on the top of the bone surface. Finally, a metal plate with a larger hole was screwed to fix both bone and silicone rubber within the mold. The material for the bone adhesive was applied in the resulting hole of the silicone matrix and subsequently cured with the LED lamp four times for 30 s as mentioned above. A static mechanical testing machine Zwick 1440 with a 2.5 kN load cell was used to determine the shear strength of the bone adhesive joints after 1 h dry storage as well as after 1 and 7 d storage in PBS at 37 $^{\circ}$ C. The test settings were 1 N for the initial preload and a cross-head speed of 1 mm min⁻¹. A scheme of this setup is shown in Figure S5B (Supporting Information).

Characterization Techniques: To analyze the conversion degree of the polymerization and the polyaddition reaction, a FT-IR spectrometer with attenuated total reflectance (ATR) was used (Nicolet iS10, Fisher Scientific, Waltham, MA, USA). The degree of conversion was calculated on the basis of peak area changes of the methacrylic bond at 1637 cm⁻¹ normalized to the peak area of C=O bond at 1720 cm⁻¹ before and after light curing.^[18,43] The depth of curing was measured by filling a polyethylene cylindrical mold (d = 6 mm, h = 12 mm) with the adhesive followed by 60 s blue light irradiation from one side. The partially cured adhesive was removed from the mold and cleaned by rinsing with acetone. The thickness of the resulting part was measured with a digital caliper. X-ray diffraction patterns of crosslinked bone adhesives before and after 7 d aging in PBS were recorded on a diffractometer D5005 (Siemens, Karlsruhe, Germany) and the phase composition was checked by means of Joint Committee on Powder Diffraction Standards (JCPDS) reference patterns. Stereomicroscopy (SteREO Discovery.V20, Carl Zeiss, Jena, Germany) was used to analyze the fracture surface, and surface morphology and elemental composition was revealed by SEM (Focused

Ion Beam (FIB)-SEM-microscope CB 340, Carl Zeiss, Jena, Germany) and EDX to identify bone adhesive residues on the surface.

In Vitro Cytocompatibility: Mouse fibroblasts L929 CC1 (ATCC, Wesel, Germany) were cultured in Dulbecco's modified Eagle's medium (DMEM) supplemented with 1% penicillin and streptomycin, 1% 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES) buffer (1 M solution) and 10% fetal calf serum (FCS). Human fetal osteoblast cell line hFOB 1.19 (LGC Standards, Wesel, Germany) was cultured in DMEM supplemented with 1% penicillin and streptomycin, 0.3 mg mL⁻¹ geneticin (G-418 sulfate) and 10% FCS. All components were purchased from Invitrogen Life Technologies, Karlsruhe, Germany. Incubation of L929 was performed according to the manufacturer's instructions in a humidified 5% CO₂ incubator at 37 °C, for hFOB 1.19 cell line a humidified 5% CO₂ incubator at 34 °C was used.

For cytocompatibility measurements according to DIN EN ISO 10993-5 and 10993-12, disk-shaped samples (15 mm diameter × 2 mm height) were prepared from the bone adhesives as described above both with newberyite filler, 0.1 wt% TEMED and 20 wt% NCO-sP(EO-stat-PO) as well as pure hydrogels from 20 wt% NCO-sP(EO-stat-PO) in water. For the cell test with hFOB 1.19 cell line, bone adhesives with newberyite filler and different amounts of TEMED (0 and 0.1 wt%) without and with 20 wt% NCO-sP(EO-stat-PO) were prepared. All specimens were placed in quadruplicate in a 24-well plate (Nunc, Wiesbaden, Germany) and washed three times with sterile phosphate buffered saline (PBS), sterilized with 1 mL ethanol for 5 min and again washed two times with sterile PBS. After incubation of the samples in DMEM for 48 h, the eluate was collected and used in further cell experiments without dilution. Cells were seeded in triplicate in a 48-well plate (Nunc, Wiesbaden, Germany) with an initial density of 50 000 cells cm⁻². As positive control, Vekaplan KT poly(vinyl chloride) (PVC) plates (König GmbH, Wendelstein, Germany) and as negative control PS surfaces of the cell culture plates were used. After adding the eluate to the cell culture for 48 h, the L929 cell number was examined by means of a cell counter CASY 1 (Roche, Mannheim, Germany). hFOB 1.19 cell numbers were calculated after cultivation for 3, 7, and 10 d. Respectively, for determination of cell activity, the cell proliferation reagent WST-1 (Roche Diagnostics, Mannheim, Germany) was used. After incubation of the cells at 37 °C or 34 °C and 5% CO₂ for 30 min with WST-1 mixed with cell culture medium in a ratio of 1:10, the absorption of the supernatant was measured by a Tecan spectra fluor plus photometer (Tecan, Crailsheim, Germany) at 450 nm. Both cell activity and cell number were normalized to PS as reference.

Analysis of variance (ANOVA) was examined by means of SigmaPlot (Systat Software GmbH, Erkrath, Germany) using one-way, two-way, or three-way ANOVA depending on the raw data. In order to calculate the statistical significance, a post hoc Tukey test was performed.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

The authors gratefully acknowledge financial support by the "Deutsche Forschungsgemeinschaft" (DFG) under Grant Nos. GB1/20-1 and GR3232/3-1 and the DFG State Major Instrumentation Programme, funding the crossbeam scanning electron microscope Zeiss CB 340 (INST 105022/58-1 FUGG).

Received: August 12, 2016

Revised: November 2, 2016

Published online:

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